

The Special Composition Question

February 24th, 2004

Mereological terms and definitions

- Mereology is the theory of parts and wholes.
 - Confusingly, ‘Mereology’ is also the name of a particular, controversial theory about parts and wholes: I’ll talk about it later.
- The distinctive vocabulary of that theory can all be defined in terms of the relation *being part of*
 - Standardly understood in such a way that everything is part of itself.

- ‘ x overlaps y ’ is defined as ‘there is a z which is part of x and part of y ’
- ‘the x s are disjoint’ is defined as ‘no two of the x s overlap’
- ‘ y is a fusion of the x s’ (or ‘sum’) is defined as ‘each of the x s is part of y , and everything that overlaps y overlaps at least one of the x s’
- ‘the x s compose y ’ is defined as ‘ y is a fusion of the x s, and the x s are disjoint’

- ‘ x is a proper part of y ’ is defined as ‘ x is part of y , and x is not identical to y ’
- ‘ x is simple’ is defined as ‘ x has no proper parts’
- ‘ x is composite’ is defined as ‘ x is not simple’

- Note: for van Inwagen and many others, the notion of parthood is relativised to *times*. So the real primitive is ‘ x is part of y at t ’; and all the defined notions are time-relative in the same way.

The Special Composition Question

- ‘When is it true that there is something that the x s compose?’
- A ‘practical’ version: ‘Suppose one had certain nonoverlapping objects, the x s, at one’s disposal; what would one have to do—what *could* one do—to get the x s to compose something?’

Answers to the Special Composition Question

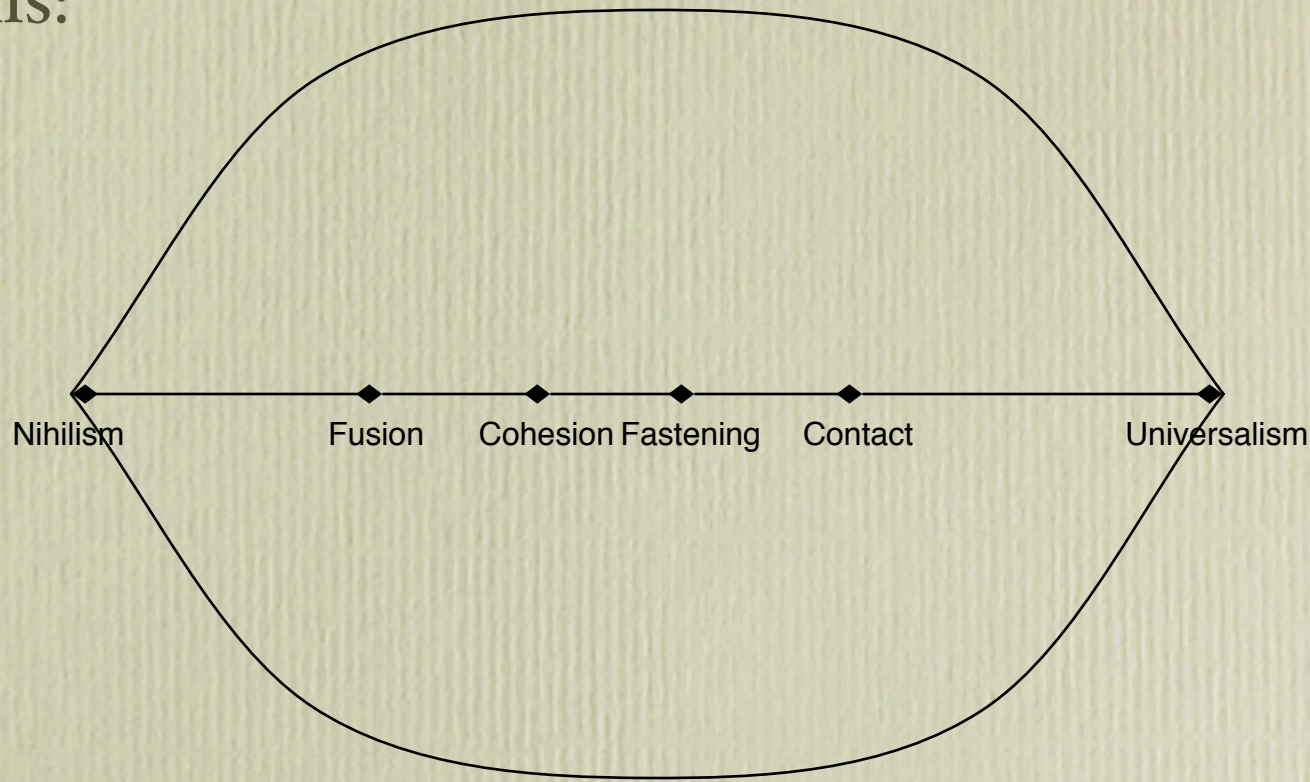
- *Contact:*
 - ‘To get the x s to compose something, one need only bring them into contact; if the x s are in contact, they compose something; and if they are not in contact, they do not compose anything.’
 - ‘The x s are in contact if (1) no two of them overlap spatially, and (2) if y and z are among the x s, then y is in contact with z , or y is in contact with w , which is one of the x s, and w is in contact with z --and so on.’
 - What does ‘ x and y are in contact’ mean? Van Inwagen doesn’t offer a definition.

- *Fastening*: some things compose something if they are *fastened* to one another: such that most ways of applying force to them would not cause them to cease to be in contact.
- *Cohesion*: some things compose something if they *cohere*—if they are such that they can't be pulled apart without breaking some of them.
- *Fusion*: some things compose something if they are *fused*, 'so that they melt into each other in a way that leaves no discoverable boundary'.

- *Nihilism*: it never happens that several things compose something.
 - So everything is simple.
- *Universalism* (previously referred to as the ‘Principle of Unrestricted Composition’): any things whatsoever compose something, provided they are disjoint.

- These are not all the possible answers to the Special Composition Question!
- There are many possible answers that are much too eccentric to have names.
 - EG: ‘the x s compose something if and only if no two of them are less than 17 miles apart’

- We can picture the possible answers to the special composition as laid out in a space, like this:



More eccentric views are located near the top and bottom of the map.